

Workshop 8 TEMPLATE

12pm

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In this workshop, we will answer the question: How do specific leaf area, water treatment, and species influence plant mass?

Specific leaf area is a continuous variable measured in cm^2/g .

Water treatment is a categorical variable (i.e. a *factor*) with 2 levels: drought stressed (DS) and well watered (WW).

Species is a categorical variable (again, a *factor*) with 6 levels.

Species name	Species code	Common name
<i>Encelia californica</i>	ENCCAL	Bush sunflower
<i>Eschsholzia californica</i>	ESCCAL	California poppy
<i>Penstemon centranthifolius</i>	PENCEN	Scarlet bugler
<i>Grindelia camporum</i>	GRINCAM	Gumweed
<i>Salvia leucophylla</i>	SALLEU	Purple sage
<i>Stipa pulchra</i>	STIPUL	Purple needlegrass
<i>Lotus scoparius</i>	LOTSCO	Deerweed

1. Set up

```
# read in your packages here
library(tidyverse) # general use
library(here) # file organization
library(janitor) # cleaning data frames
library(readxl) # reading excel files
library(scales) # modifying axis labels
library(ggeffects) # getting model predictions
library(MuMin) # model selection

# read in your data here

drought_exp <- read_xlsx(
  here("data", "Valliere_etal_EcoApps_Data.xlsx"),
  sheet = "First Harvest"
)
```

```
# storing colors to use for species
lotsco_col <- "#E69512"
pencen_col <- "#D6264F"
salleu_col <- "#6D397D"
enccal_col <- "#3A5565"
stipul_col <- "#3F564F"
```

```

escscal_col <- "#515481"
gricam_col <- "#6C91BD"

# storing colors to use for water treatments
ds_col <- "#A62F03"
ww_col <- "#045CB4"

# storing a ggplot theme (that will be used for all ggplots)
theme_set(theme_bw())

```

2. Clean data

Data source: Valliere, Justin; Zhang, Jacqueline; Sharifi, M.; Rundel, Philip (2019). Data from: Can we condition native plants to increase drought tolerance and improve restoration success? [Dataset]. Dryad. <https://doi.org/10.5061/dryad.v0861f7>

```

# creating new object for clean data
drought_exp_clean <- drought_exp |>
  # cleaning names to make legible
  # adding underscores and replaces capitals
  clean_names() |>
  # creating new column with scientific name
  mutate(species_name = case_match(
    species,
    "ENCCAL" ~ "Encelia californica", # bush sunflower
    "ESCCAL" ~ "Eschscholzia californica", # California poppy
    "PENCEN" ~ "Penstemon centranthifolius", # Scarlet bugler
    "GRICAM" ~ "Grindelia camporum", # Gumweed
    "SALLEU" ~ "Salvia leucophylla", # purple sage
    "STIPUL" ~ "Stipa pulchra", # purple needlegrass
    "LOTSCO" ~ "Lotus scoparius" # deerweed
  )) |>
  # Creating new column with treatment names
  mutate(water_treatment = case_match(
    water,
    "WW" ~ "Well watered",
    "DS" ~ "Drought stressed"
  )) |>
  # reordering species' names
  mutate(species_name = as_factor(species_name),
         species_name = fct_relevel(species_name,

```

```

        "Lotus scoparius",
        "Penstemon centranthifolius",
        "Salvia leucophylla",
        "Encelia californica",
        "Stipa pulchra",
        "Eschscholzia californica",
        "Grindelia camporum")) |>

# reordering treatment levels
mutate(water_treatment = as_factor(water_treatment),
       water_treatment = fct_relevel(water_treatment,
                                     "Drought stressed",
                                     "Well watered")) |>

# selecting certain columns to keep/ making data concise
select(species_name, water_treatment, sla, total_g)

```

Insert code to double check the structure of the data frame here.

Insert code to display 10 random rows from the data frame here.

```

slice_sample(
  drought_exp_clean, # clean data fram to display
  n = 10 # show ten rows
)

```

A tibble: 10 x 4

	species_name <fct>	water_treatment <fct>	sla <dbl>	total_g <dbl>
1	Eschscholzia californica	Well watered	122.	0.541
2	Penstemon centranthifolius	Drought stressed	146.	0.126
3	Penstemon centranthifolius	Drought stressed	143.	0.0881
4	Salvia leucophylla	Drought stressed	77.6	0.186
5	Stipa pulchra	Drought stressed	146.	0.246
6	Lotus scoparius	Drought stressed	99.8	0.0734
7	Lotus scoparius	Drought stressed	91.4	0.0626
8	Salvia leucophylla	Well watered	127.	0.258
9	Stipa pulchra	Well watered	163.	0.370
10	Encelia californica	Drought stressed	161.	0.273

3. Visualizing data

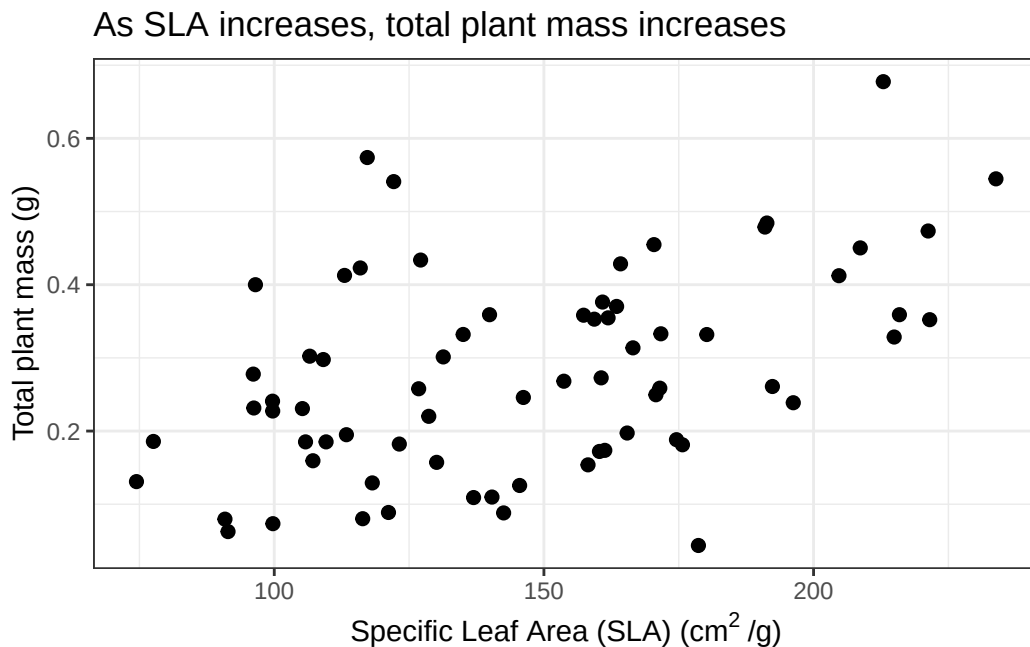
What is the relationship between SLA and plant mass?

Make a scatterplot to visualize SLA (the predictor) on the x-axis, and total mass (the response) on the y-axis.

Label the axes with units where appropriate.

Add a title describing the relationship (e.g. as SLA increases, total mass _____).

```
#ggplot base layer, drought clean
ggplot(data = drought_exp_clean,
       # x axis = sla
       # y axis= total mass
       mapping = aes(x = sla,
                     y = total_g)
) +
  #adding points
  geom_point(size = 2) +
  # labelling axes with units and adding a title
  labs(y = "Total plant mass (g)",
       x = "Specific Leaf Area (SLA) (cm2/g)",
       title = "As SLA increases, total plant mass increases")
```



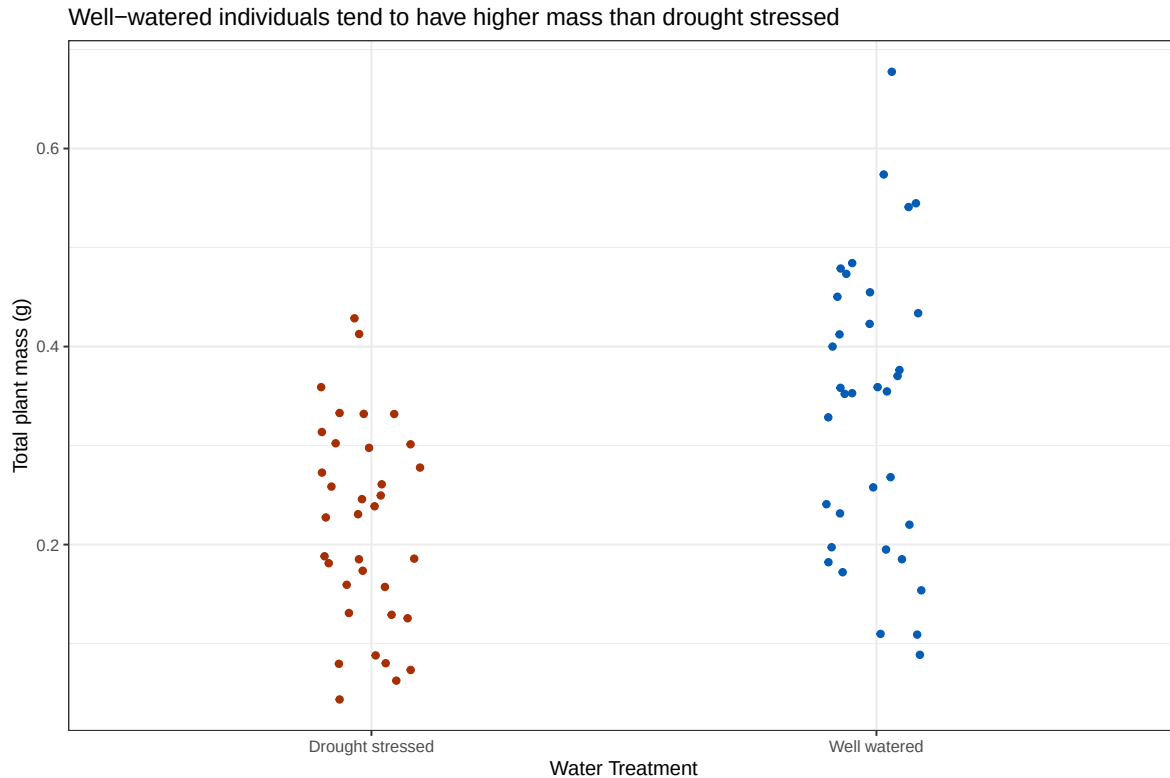
What are the differences in total mass between water treatments?

Make a jitterplot to visualize the differences in plant mass between water treatments.

Label the axes with units where appropriate.

Add a title describing the differences in (mean or median) plant mass between water treatments.

```
ggplot(data = drought_exp_clean,
  # x axis = water treatment
  # y axis= total mass
  mapping = aes(x = water_treatment,
    y = total_g,
    color = water_treatment)
) +
  #adding points
  geom_jitter(height = 0,
    width = 0.1
  ) +
  scale_color_manual(values = c(
    "Drought stressed" = ds_col,
    "Well watered" = ww_col
  )) +
  # labelling axes with units and adding a title
  labs( y = "Total plant mass (g)",
    x = "Water Treatment",
    title = "Well-watered individuals tend to have higher mass than drought stressed") +
  theme(legend.position = "none")
```



What are the differences in total mass between species?

Make a jitterplot to visualize the differences in plant mass between species.

Label the axes with units where appropriate.

Add a title describing the differences in (mean or median) plant mass between species.

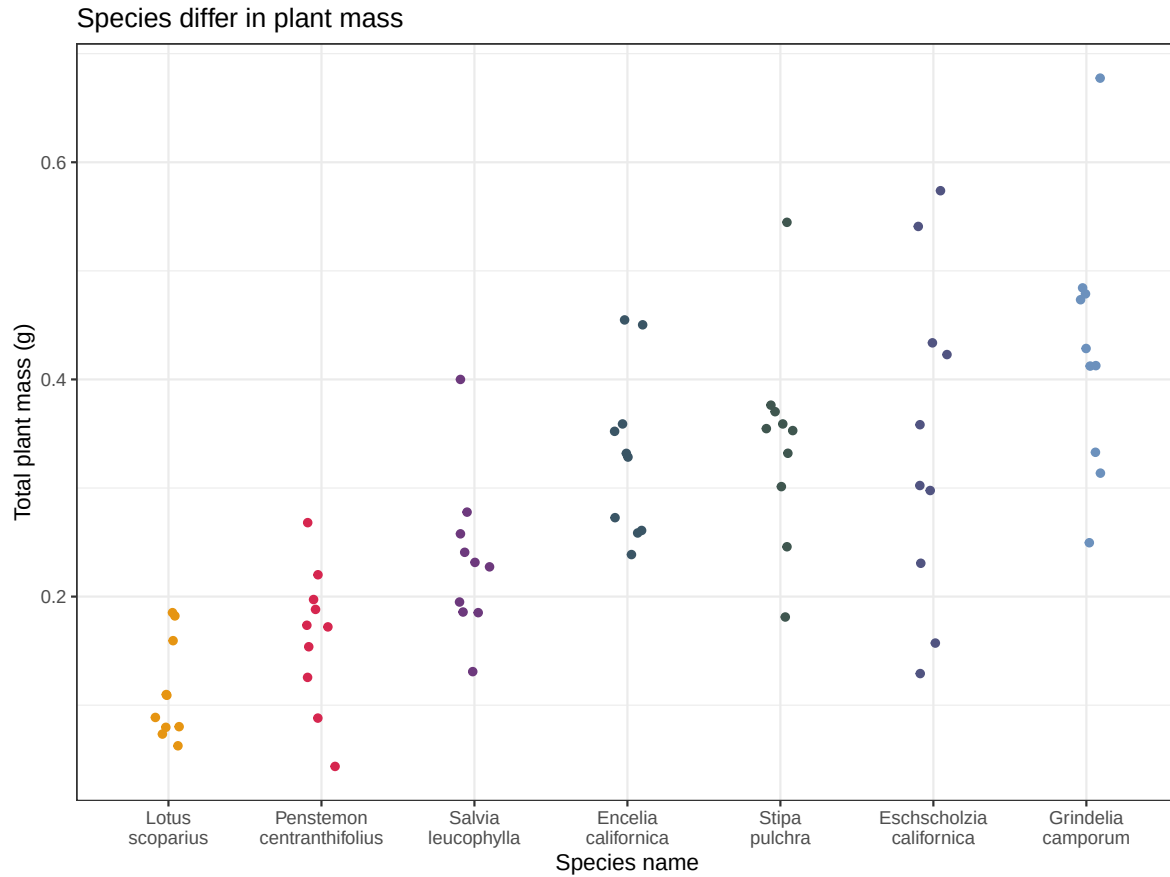
Bonus: order the x-axis by decreasing mean plant mass.

```
ggplot(data = drought_exp_clean,
  # x axis = species
  # y axis= total mass
  mapping = aes(x = species_name,
    y = total_g,
    color = species_name)
) +
  #adding points
  geom_jitter(height = 0,
    width = 0.1
```

```

    ) +
#wrapping x axis names
scale_x_discrete(labels = label_wrap(10)) +
# labelling axes with units and adding a title
labs( y = "Total plant mass (g)",
      x = "Species name",
      title = "Species differ in plant mass") +
#custom colors
scale_color_manual(
  values = c(
    "Lotus scoparius" = lotsco_col,
    "Penstemon centranthifolius" = pencen_col,
    "Salvia leucophylla" = sallev_col,
    "Encelia californica" = enccal_col,
    "Stipa pulchra" = stipul_col,
    "Eschscholzia californica" = esccal_col,
    "Grindelia camporum" = gricam_col
  )
) +
theme(legend.position = "none")

```

4. Fitting models

8 models total:

Model number	SLA	Water treatment	Species	Predictor list
0				no predictors (null model)
1	X	X	X	all predictors (full model)
2	X	X		SLA and water treatment
3	X		X	SLA and species
4		X	X	water treatment and species
5	X			SLA
6		X		water treatment
7			X	species

Model fitting

```
# model 0: null model

model0 <- lm(
  total_g ~ 1,
  data = drought_exp_clean
)

# model 1: all predictors

model1 <- lm(
  total_g ~ sla + water_treatment + species_name,
  data = drought_exp_clean
)

# model 2: SLA and water treatment

model2 <- lm(
  total_g ~ sla + water_treatment,
  data = drought_exp_clean
)

# model 3: SLA and species

model3 <- lm(
  total_g ~ sla + species_name,
  data = drought_exp_clean
)

# model 4: water treatment and species

model4 <- lm(
  total_g ~ species_name + water_treatment,
  data = drought_exp_clean
)

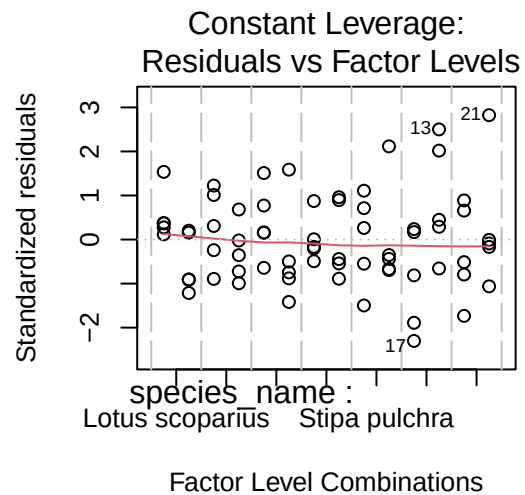
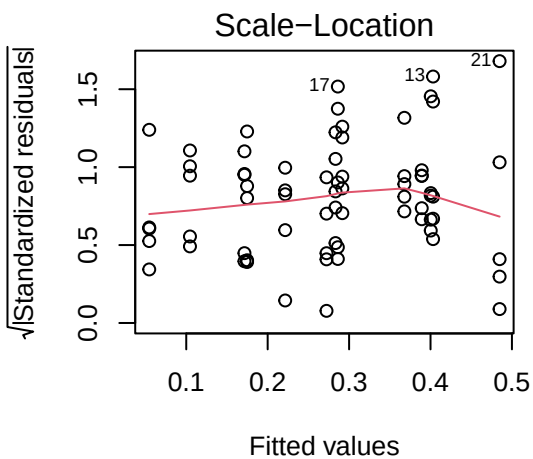
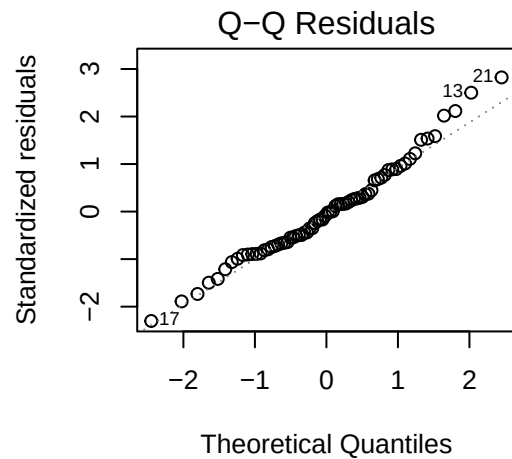
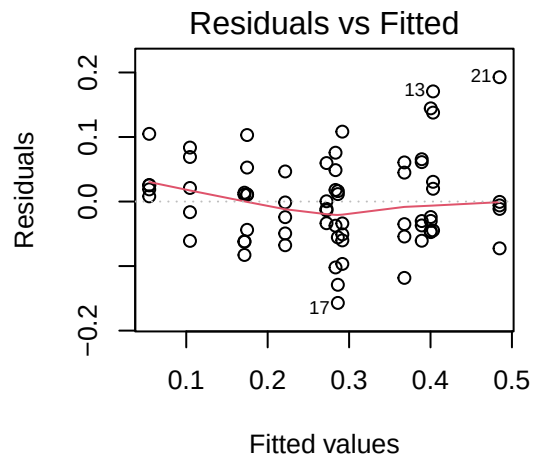
# model 5: SLA

model5 <- lm(
  total_g ~ sla,
  data = drought_exp_clean
)
```

```
)  
  
# model 6: water treatment  
  
model6 <- lm(  
total_g ~ water_treatment,  
data = drought_exp_clean  
)  
  
# model 7: species  
  
model7 <- lm(  
total_g ~ species_name,  
data = drought_exp_clean  
)
```

Model diagnostics

```
#displaying plots  
par(mfrow = c(2,2))  
#plotting diagnostics  
plot(model4)
```



- residuals are normally distributed (QQ plot)
- residuals are homoscedastic (residuals and scale-location)
- no outliers that could be influencing model coefficient estimates (leverage)

Model selection

Insert code to use the `AICc()` function to calculate the AIC for each model.

Arrange by decreasing AIC.

```
AICc(
model0,
model1,
model2,
model3,
model4,
model5,
model6,
model7
) |>
arrange(AICc)
```

	df	AICc
model4	9	-156.19595
model11	10	-153.75361
model13	9	-124.07569
model17	8	-120.30191
model2	4	-95.82521
model5	3	-88.85180
model6	3	-86.76661
model0	2	-74.98036

Insert code to display the summary for the best model.

Stop and think: what does this model mean?

What is the best model?

The best model includes water treatment and species as predictors

How much variation in the response (total mass, in grams) does this model explain?

72.6% of variation in plant mass is explained by this model

How do we interpret the effects of the predictors on the response variable (again, total mass in grams)?

- Species influences plant mass
- water treatment influences plant mass

- ‘(Intercept)’: The intercept is the model prediction for plant mass for *Lotus scoparius* in the drought stressed treatment (*lotus scoparius* and drought stressed were the reference levels in the summary)
- ‘water_treatmentWell waters’: model prediction for dif in *lotus scoparius* mass between well watered and drought stressed treatments.
- ‘species_namePenstemon centranthifolius’: model prediction for dif in *penstemon centranthifolius* mass between well watered and drought stressed treatments

4. Model predictions

Insert code to generate model predictions for the best model.

Rename the `x` column and `group` column to match the original data frame (`drought_exp_clean`).

Look at the object before plotting.

```
preds <- ggpredict(
  model4,
  terms = c("water_treatment", "species_name")
) |>
  rename(water_treatment = x,
         species_name = group)
```

5. Final figure

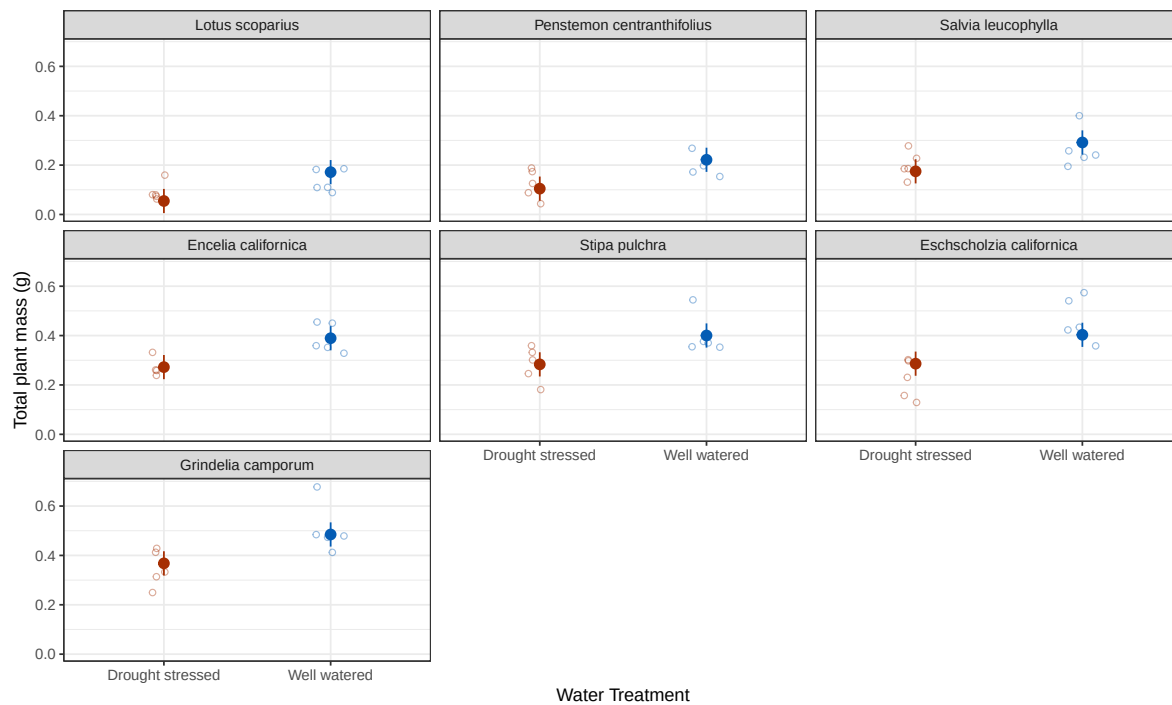
Create a clean, publication ready figure depicting predictions for the best model and the underlying data.

```
ggplot(data = drought_exp_clean,
       mapping = aes(x = water_treatment,
                     y = total_g,
                     color = water_treatment)) +
  # individual plants for each treatment
  geom_jitter(height = 0,
             width = 0.1,
             alpha = 0.4,
             shape = 21) +
  # predictions and confidence intervals
  geom_pointrange(data = preds,
```

```

mapping = aes(x = water_treatment,
              y = predicted,
              ymin = conf.low,
              ymax = conf.high)) +
#creates different panels for each series
facet_wrap(~ species_name
           ) +
# custom colors
scale_color_manual(values = c(
  "Drought stressed" = ds_col,
  "Well watered" = ww_col
)) +
theme(legend.position = "none") +
labs(x = "Water Treatment",
     y = "Total plant mass (g)")

```



6. Writing

Figure caption

Basic components of test

predictors

response

test

distribution

parameters (degrees of freedom)

R^2

p-value

significance level

reference level(s)

Written communication

Methods

We used linear models to determine the predictive relationship between specific leaf area (cm^2/g), treatment, and species on plant mass (g). We evaluated model assumptions of normally distributed errors using QQ plots of model residuals. We evaluated assumptions of constant variance of errors using Residuals vs Fitted and Scale-Location plots. We used Akaike's Information Criterion (AIC) to select the simplest model that explained the most variation in plant mass.

Results

We found that the simplest model describing the greatest variation in plant mass included treatment and species as predictors (linear model: $F(2, 67) = 14.57$, $p < 0.001$, $\alpha = 0.05$).